

[0:00–0:15]

Introduction: Why This Matters

[Visual: Title slide “Adaptive Discharge AI — Safer, Simpler Discharge Instructions”]

Voiceover:

Hello. Too often, patients leave the hospital with discharge instructions written at a complex medical level — sometimes requiring a college education just to understand. The result? Confusion, medication errors, and unnecessary readmissions.

[0:15–0:30]

The Solution: Who Benefits and What We Do

[Visual: Slide showing two persons: "Patient with Low Health Literacy" and "Older Patient"]

Voiceover:

Our solution is Adaptive Discharge AI. It helps patients with lower health literacy or elderly individuals by rewriting complex instructions into accessible language tailored to their profile. Best of all — it does so without altering critical numbers or facts.

[0:30–0:45]

Technology: Models and Why MedGemma

[Visual: MedGemma logo + text “Open-weights, healthcare-tuned”]

Voiceover:

At its core is MedGemma from HAI-DEF. We chose this model because it has open weights and is specifically tuned for healthcare. It’s much better suited than general language models and essential for hospital environments where data privacy is critical.

[0:45–1:45]

Live Demo (Screen Capture)

[Visual: Streamlit interface. Cursor inputs a complex sample text with doses, times, and emergency numbers on the left.]

Voiceover:

Let’s see it in action. On the left, we input standard, complex medical instructions with important numbers and deadlines.

[Visual: Select “low\_literacy” profile. Click “Simplify”. The simplified output appears, organized into clear sections.]

Voiceover:

We select the low literacy profile and generate a simplified version. The output is structured into sections: Medications, Follow-up Actions, Safety.

[Visual: Open the “Quality & Safety Checks” panel. Emphasize FKGL readability index dropping and the green checkmark “Numbers/critical words preserved: ”.]

Voiceover:

The most crucial part is here in the safety panel. We see the readability index improves significantly, and the system confirms all numbers and key words are maintained at 100%.

[Visual: Quick glimpse of the interface.]

Voiceover:

We use strict prompting and post-validation. If the model misses a fact, we retry. If the risk remains, we revert to the original for safety.

[Visual: Switch profile to “elderly” or “post\_surgery” and show a different output.]

Voiceover:

You can quickly switch profiles — for example, to “elderly” — and the tone of instructions adapts instantly.

[1:45–2:10]

How It Works (Agent Flow)

[Visual: Diagram of the process: Input → Section extractor → Profile adapter → MedGemma → Guardrails → Output]

Voiceover:

The architecture follows a multi-step agent flow: first, sections are extracted and divided. Then, they’re adapted to the profile. Next, MedGemma rewrites the text. Crucially, our guardrails lock in numbers and key terms. Finally, we show the validated result.

[2:10–2:35]

Metrics and Impact

[Visual: Simple table showing validation results — FKGL reduced by 3–4 points, 100% preservation of numbers]

Voiceover:

Validation results show a significant increase in readability — 3 to 4 points lower FKGL — while guaranteeing 100% fact preservation. This isn’t just numbers — it’s a direct path to better adherence and fewer incidents at home.

[2:35–2:55]

Deployment and Practice

[Visual: Slide with “Feasibility” and “Edge (future)” options]

Voiceover:

Today, we operate in a cloud environment with 4-bit quantization, but the architecture is fully ready for on-prem deployment in hospitals. Security measures can also run locally for maximum safety.

[2:55–3:00]

Final Slide

[Visual: Links to repo, demo, and documentation]

Voiceover:

This is Adaptive Discharge AI — clearer instructions for safer discharge. Thank you for watching.